MYP 5 CHEMISTRY

LAB REPORT GUIDE

Mr. Mercado

Name:	Lab:	Date:
________________________	________________________	________________

HOW TO USE THIS GUIDE
Use this guide while writing your laboratory report by hand. For each section: 1) Read What to Include. 2) Follow the suggested structure. 3) Look at the chemistry example. 4) Check your work before moving to the next section.

1. RESEARCH QUESTION

The Research Question clearly states what you are investigating.

A strong Research Question identifies:

- the Independent Variable (IV)
- the Dependent Variable (DV)
- the relationship being investigated between them

Independent Variable - IV

The variable that is intentionally changed during the experiment.

ASK YOURSELF
What am I changing?

Dependent Variable - DV

The variable that is measured or observed as a result.

ASK YOURSELF
What am I measuring or observing?

Recommended Format

FORMAT
How does [Independent Variable] affect [Dependent Variable]?

Chemistry Example

EXAMPLE
How does water temperature affect the time required for a sugar cube to dissolve in water?

Independent Variable (IV): Water temperature (°C)

Dependent Variable (DV): Time required for the sugar cube to dissolve (s)

Check Your Research Question

- The investigation is clearly stated.
- The IV is identified.
- The DV is identified.
- The relationship between the IV and DV is clear.

2. VARIABLES

Variables are the factors in an investigation that are changed, measured, or kept the same.
A well-designed investigation should clearly identify:

- Independent Variable
- Dependent Variable
- Controlled Variables

Independent Variable - IV

The Independent Variable is the factor that you intentionally change during the experiment.

ASK YOURSELF
What am I changing?

You should also identify the different values or variations of the Independent Variable used in the investigation.

CHEMISTRY EXAMPLE
Independent Variable: Water temperature (°C) Variations tested: 20 °C, 40 °C, 60 °C

Dependent Variable - DV

The Dependent Variable is the factor that you measure or observe as a result of changing the Independent Variable.

ASK YOURSELF
What am I measuring or observing?

Always include the appropriate scientific unit when the variable is quantitative.

CHEMISTRY EXAMPLE
Dependent Variable: Time required for the sugar cube to completely dissolve (s) The dissolving time will be measured using a stopwatch and recorded in seconds.

Controlled Variables

Controlled Variables are the factors that must be kept the same during every trial.
Keeping these variables constant helps make the investigation a fair test.

ASK YOURSELF
What things must stay the same so that only the Independent Variable affects the results?

Controlled Variable	How It Will Be Controlled
Amount of water	Use 100 mL of water for every trial
Sugar cube size	Use the same type and size of sugar cube for every trial
Container	Use the same size and type of beaker
Stirring	Do not stir the solution during any trial
Measurement method	Use the same stopwatch and method for every trial

3. HYPOTHESIS

A Hypothesis is a scientifically supported prediction about what you think will happen during the investigation.

Your hypothesis should predict the relationship between the Independent Variable and the Dependent Variable.

A strong hypothesis also explains why you expect the result.

Recommended Format

FORMAT
If [Independent Variable] is changed, then [Dependent Variable] will __________ because __________.

Chemistry Example

EXAMPLE
If the temperature of the water increases, then the time required for the sugar cube to dissolve will decrease because particles move faster at higher temperatures and interact more frequently with the sugar particles.

Break It Down

IF - What are you changing?

If the temperature of the water increases...

THEN - What do you predict will happen to the Dependent Variable?

...then the time required for the sugar cube to dissolve will decrease...

BECAUSE - What is the scientific reason for your prediction?

...because particles move faster at higher temperatures and interact more frequently with the sugar particles.

Weak Hypothesis

WEAK EXAMPLE
I think the sugar will dissolve faster in hot water.

This gives a prediction, but it does not clearly identify the relationship between the variables or explain the science.

Strong Hypothesis

STRONG EXAMPLE
If the temperature of the water increases, then the dissolving time of the sugar cube will decrease because particles at higher temperatures have greater kinetic energy and move more rapidly.

4. BACKGROUND INFORMATION

The Background Information explains the science needed to understand your investigation.

Use the class textbook as your main source of scientific information.

Your Background Information should contain two paragraphs.

Paragraph 1 - Purpose of the Investigation

- What are you investigating?
- What is the purpose of the experiment?
- Why are you collecting this information?

SENTENCE STARTER
The purpose of this investigation is to determine how __________ affects __________.

CHEMISTRY EXAMPLE
The purpose of this investigation is to determine how water temperature affects the amount of time required for a sugar cube to dissolve. Different water temperatures will be tested and the dissolving time will be measured.

Paragraph 2 - Scientific Information About the IV, DV, and Their Relationship

- What is my IV?
- What does it mean scientifically?
- What important information should someone know about it?

CHEMISTRY EXAMPLE
Temperature is a measure related to the average kinetic energy of particles in a substance. When temperature increases, particles generally move faster and have greater kinetic energy.

Dependent Variable (DV)

- What is my DV?
- What exactly am I measuring?
- What scientific process does it represent?

CHEMISTRY EXAMPLE
Dissolving occurs when particles of a solute become separated and distributed throughout a solvent. In this investigation, the time required for the sugar cube to completely dissolve is measured in seconds.

Relationship Between the IV and DV

- Explain scientifically how or why the IV may affect the DV.
- Connect the scientific information about the IV and DV.

SENTENCE STARTER
As __________ changes, __________ may change because __________.

CHEMISTRY EXAMPLE
As water temperature increases, the sugar cube may dissolve in less time. At higher temperatures, water particles move faster and interact more frequently with the sugar particles, which may increase the rate at which the sugar dissolves.

5. MATERIALS

List everything that was actually used during the investigation.

Your materials should include:

- materials
- equipment
- quantities
- sizes, amounts, or measurements
- appropriate scientific units

Be specific.

Weak Example

WEAK EXAMPLE
Water Beakers Sugar Thermometer

This does not provide enough information.

Better Chemistry Example

- 3 x 250 mL beakers
- 300 mL of water
- 9 sugar cubes
- 1 thermometer (°C)
- 1 graduated cylinder (100 mL)
- 1 stopwatch (s)
- 1 stirring rod

6. PROCEDURE

The Procedure explains exactly how the investigation was completed.
Someone who reads your procedure should be able to understand what you did and repeat the experiment.
Your procedure should be:

- written in numbered steps
- clear
- organized
- specific
- written in the correct order
- detailed enough to repeat the investigation

Your Procedure Should Explain

- how the equipment was set up
- how the Independent Variable was changed
- how the Dependent Variable was measured
- how Controlled Variables were kept constant
- how many trials were completed
- when measurements were taken
- important safety precautions when necessary

Chemistry Example

1. Measure 100 mL of water using a graduated cylinder.
2. Pour the water into a 250 mL beaker.
3. Measure the temperature of the water using a thermometer.
4. Adjust the water temperature until it reaches approximately 20 °C.
5. Place one sugar cube into the water.
6. Start the stopwatch immediately when the sugar cube enters the water.
7. Do not stir the solution.
8. Stop the stopwatch when the sugar cube has completely dissolved.
9. Record the dissolving time in seconds in the Raw Data table.
10. Repeat the procedure two more times at 20 °C for a total of three trials.
11. Repeat Steps 1-10 using water temperatures of 40 °C and 60 °C.
12. Record all measurements and observations in the Raw Data table.

Notice What the Procedure Includes

Part	Example
Independent Variable	Temperature changes: 20 °C -> 40 °C -> 60 °C
Dependent Variable	Dissolving time is measured in seconds (s)
Controlled Variables	100 mL water, same sugar cube type, same beaker size, no stirring, same measurement method
Trials	Each temperature is tested three times

Weak Procedure

WEAK EXAMPLE
1. Get water. 2. Put sugar in it. 3. Time it. 4. Repeat.

This procedure does not provide enough information for another person to repeat the investigation accurately.

Better Procedure Writing

BETTER EXAMPLE
Instead of: Put some water in the beaker. Write: Measure 100 mL of water using a graduated cylinder and transfer it into a 250 mL beaker. Instead of: Time the reaction. Write: Start the stopwatch immediately after the sugar cube is placed in the water and stop it when the sugar cube has completely dissolved.

Procedure Writing Tips

Use Numbered Steps: Write 1, 2, 3, 4... Do not write the procedure as one large paragraph.
Include Measurements: Use specific values such as 25 mL, 5.0 g, 30 °C, or 60 s.
Include Scientific Units: Always include appropriate units with measurements.
Explain Repeated Trials: If the investigation has three trials, clearly state when and how the trials are repeated.
Keep the Procedure in Order: Write each step in the exact sequence in which it should be performed.

Safety

If the investigation involves heat, glassware, chemicals, flames, or potentially irritating substances, include appropriate safety precautions.

SAFETY EXAMPLE
Wear safety goggles throughout the investigation and handle hot glassware using appropriate laboratory equipment.

7. RAW DATA

Raw Data are the original measurements or observations collected directly during the experiment.

Raw Data have not been averaged, calculated, classified, or interpreted yet.

Your Raw Data Table Must Have

13. A descriptive title
14. The Independent Variable in the first column
15. The IV organized in ascending or logical order
16. Clearly labeled headings
17. Appropriate units
18. Clearly identified trials
19. All measurements or observations recorded

Example of Quantitative Raw Data

Table 1. Effect of Water Temperature on Sugar Cube Dissolving Time

Water Temperature (°C)	Trial 1 Time (s)	Trial 2 Time (s)	Trial 3 Time (s)
20	180	175	184
40	118	121	115
60	72	68	70

Notice

- The Independent Variable is in the first column.
- The temperatures are organized from 20 -> 40 -> 60 °C.
- Each trial has its own column.
- The unit seconds (s) is included in the heading.

Qualitative Raw Data

Not every investigation produces only numbers. You may also record qualitative observations.

- color change
- formation of bubbles
- formation of a precipitate
- odor
- change in appearance
- change in state

Example of Qualitative Data

Sample	Observation Before	Observation After
A	Clear, colorless liquid	White solid formed
B	Blue solution	Solution became lighter blue
C	White solid	Bubbles formed after mixing

Record what you observe, not what you think happened.

OBSERVATION
Bubbles formed immediately after mixing.

INTERPRETATION
A gas-producing chemical reaction occurred.

The first statement is an observation. The second statement is an interpretation.

8. PROCESSED DATA

Processed Data are created when you transform or analyze your Raw Data.

Depending on the investigation, this may include:

- calculations
- averages
- comparisons
- classifications
- scientific interpretations of qualitative observations

Processed Data should not simply repeat the Raw Data table.

Quantitative Example

Using the Raw Data:

Water Temperature (°C)	**Trial 1 (s)**	**Trial 2 (s)**	**Trial 3 (s)**
20	180	175	184
40	118	121	115
60	72	68	70

You may calculate the average dissolving time.

Sample Calculation

FOR 20 °C
Average = (Trial 1 + Trial 2 + Trial 3) ÷ Number of Trials Average = (180 s + 175 s + 184 s) ÷ 3 Average = 179.7 s

Example Processed Data Table

Table 2. Average Sugar Cube Dissolving Time at Different Water Temperatures

Water Temperature (°C)	**Average Dissolving Time (s)**
20	179.7
40	118.0
60	70.0

Qualitative Processed Data

Qualitative data can also be processed. Instead of performing a mathematical calculation, you may:

- classify observations
- compare observations
- identify patterns
- scientifically interpret observations

RAW OBSERVATION
Bubbles formed immediately when the two substances were mixed.

PROCESSED INTERPRETATION
Gas production was observed, which can be evidence of a chemical change.

9. CONCLUSION

Use the CER Format: Claim - Evidence - Reasoning.
Your conclusion must answer the Research Question using your experimental results.

C - CLAIM

What did the investigation show?
Restate the purpose or Research Question and clearly describe the main relationship, trend, or pattern.

SENTENCE STARTERS
The results show that... As __________ increased, __________ decreased/increased. Based on the results, __________ affected __________ by...

CHEMISTRY EXAMPLE
The results show that increasing water temperature decreased the amount of time required for the sugar cube to dissolve.

E - EVIDENCE

What data support your claim?
Use at least two specific data values or observations from your investigation.
Do not simply say: The data support my claim. Show the actual evidence.

CHEMISTRY EXAMPLE
At 20 °C, the average dissolving time was 179.7 s, while at 60 °C, the average dissolving time decreased to 70.0 s.

R - REASONING

Why did this happen scientifically?
Connect your evidence to scientific ideas. Use the scientific information discussed in your Background Information.

SENTENCE STARTERS
This occurred because... Scientifically, this can be explained by... The results can be explained by...

CHEMISTRY EXAMPLE
This result can be explained by particle motion. At higher temperatures, water particles have greater kinetic energy and move faster. This increases interactions between the water and sugar particles, allowing the sugar to dissolve more quickly.

Complete CER Example

COMPLETE EXAMPLE
The results show that increasing water temperature decreased the amount of time required for a sugar cube to dissolve. At 20 °C, the average dissolving time was 179.7 s, while at 60 °C, it decreased to 70.0 s. This result can be explained by the increased motion of particles at higher temperatures. Water particles with greater kinetic energy move faster and interact with the sugar particles more frequently. Therefore, the results indicate that increasing water temperature decreases the amount of time required for sugar to dissolve.

10. EVALUATION AND IMPROVEMENT

In this section, you analyze a limitation or source of error in the investigation and explain how the investigation could be improved.

You must do more than simply name an error.

20. Identify the limitation or source of error.
21. Explain how it may have affected the results.
22. Propose a realistic improvement.
23. Explain how the improvement would improve the investigation.

Weak Example

WEAK EXAMPLE
Human error occurred. We should be more careful next time.

This is too general.

Strong Chemistry Example

Part	Example
Limitation / Source of Error	The exact moment when the sugar cube completely dissolved was determined visually by the person using the stopwatch.
Effect on the Results	Different observers may judge the exact dissolving point differently, which could cause some dissolving times to be slightly longer or shorter than the actual value.
Improvement	The same person could determine the endpoint for every trial using the same criteria.
Why This Would Improve the Investigation	Using the same observer and the same endpoint criteria would make the measurements more consistent between trials and reduce variation caused by different judgments.

Another Example

Part	Example
Limitation / Source of Error	The water temperature may have changed during the experiment.
Effect on the Results	If the water cooled while the sugar was dissolving, the actual temperature may not have remained at the intended value.
Improvement	The water temperature could be monitored throughout each trial and maintained as close as possible to the selected temperature.
Why This Would Improve the Investigation	Maintaining a more constant temperature would provide better control of the Independent Variable and make comparisons between trials more reliable.

11. REFERENCES

Include the sources used for scientific information in your report.
Your class textbook must be included in the reference list.
References must follow APA 7th edition format.

Basic Book Reference Structure

FORMAT
Author's Last Name, Initial. (Year). Title of book. Publisher.

Example Format

EXAMPLE
Smith, J. A. (2024). Introduction to chemistry. Science Press.

In-Text Citation Example

Scientific information taken from a source should also be cited in your Background Information.
Parenthetical Citation: (Smith, 2024)
Narrative Citation: Smith (2024) explains that...